ISOMORPHISMS IN THE CATEGORY OF WEIRD INVERSE SYSTEMS ARE DETECTED BY BOUNDED TORSION KERNELS AND COKERNELS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof span ends with the author's sentence 'We omit the verification that (c', psi_n) is the inverse of (c, phi_n) in the category.'; that sentence is kept verbatim and no step was supplied in its place. The single reference inside the counted statement span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- The category of weird inverse systems and its morphisms (c, phi_n). referred to at statement lines 29169-29170. Fixes the objects, the morphism datum and the meaning of isomorphism being characterized.
- Torsion submodules E[I^c] and the quotients E/E[I^{c'}]. used at proof lines 29180-29188. Provides the ambient modules in which the candidate inverse maps take values.
- Annihilation of the cokernel by I^{c'}. used at proof lines 29181-29182. Is what makes the lifting of x to I^cE_n possible.

Source reference. Original remark 29141: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29141–29141. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *A morphism (c, φ_n) of the category of Remark 29141 is an isomorphism if and only if there exists a $c' \geq 0$ such that $\mathrm{Ker}(\varphi_n)$ and $\mathrm{Coker}(\varphi_n)$ are $I^{c'}$-torsion for all $n \gg 0$.*

Proof. We may and do assume $c' \geq c$ and that the $\mathrm{Ker}(\varphi_n)$ and $\mathrm{Coker}(\varphi_n)$ are $I^{c'}$-torsion for all n. For $n \geq c'$ and $x \in I^{c'}E'_n$ we can choose $y \in I^cE_n$ with $x = \varphi_n(y) \bmod E'_n[I^c]$ as $\mathrm{Coker}(\varphi_n)$ is annihilated by $I^{c'}$. Set $\psi_n(x)$ equal to the class of y in $E_n/E_n[I^{c'}]$. For a different choice $y' \in I^cE_n$ with $x = \varphi_n(y') \bmod E'_n[I^c]$ the difference $y - y'$ maps to zero in $E'_n/E'_n[I^c]$ and hence is annihilated by $I^{c'}$

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in I^cE_n. Thus the maps $\psi_n : I^{c'}E'_n \to E_n/E_n[I^{c'}]$ are well defined. We omit the verification that (c', ψ_n) is the inverse of (c, φ_n) in the category. $\square$